2.3.1 Irrational Numbers on the Number Line

Where are the points on the number line that correspond to the irrational numbers? As an example, let us locate $\sqrt{2}$ on the number line. This is easy.

Remember that $\sqrt{2}$ is the length of the diagonal of the square whose side is 1 unit (How?)Simply construct a square and transfer the length of one of its diagonals to our number line. (see Fig.2.7).

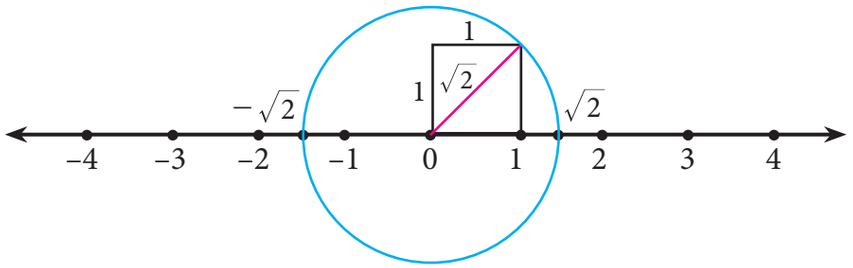


Fig.2.7

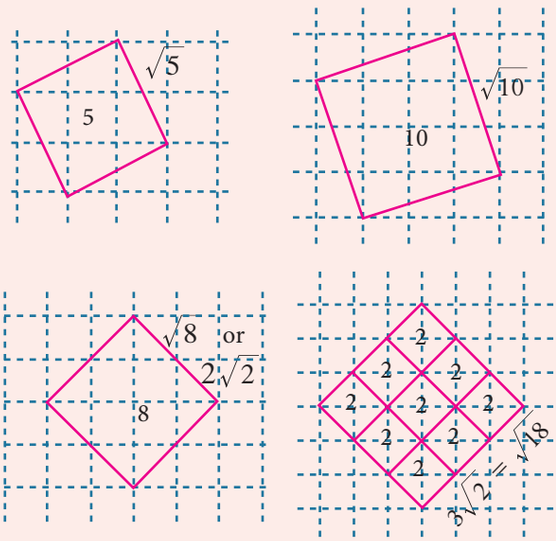
We draw a circle with centre at 0 on the number line,with a radius equal to that of diagonal of the square. This circle cuts the number line in two points, locating $\sqrt{2}$ on the right of 0 and $-\sqrt{2}$ on its left. (You wanted to locate $\sqrt{2}$; you have also got a bonus in $-\sqrt{2}$)

You started with Natural numbers and extended it to rational numbers and then irrational numbers. You may wonder if further extension on the number line waits for us. Fortunately it stops and you can learn about it in higher classes.

Representation of a Rational number as terminating and non terminating decimal helps us to understand irrational numbers. Let us see the decimal expansion of rational numbers.

Squares on grid sheets can be used to produce irrational lengths.

Here are a few examples :



2.3.2 Decimal Representation of a Rational Number

If you have a rational number written as a fraction, you get the decimal representation by long division. Study the following examples where the remainder is always zero.

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